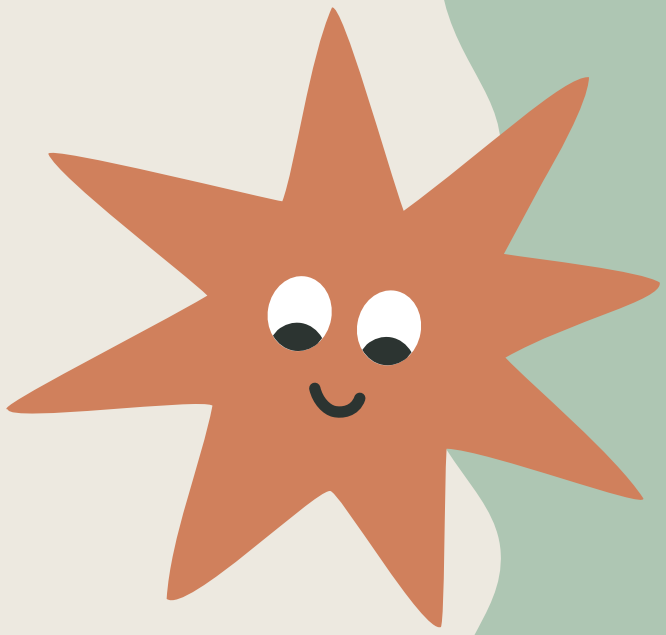
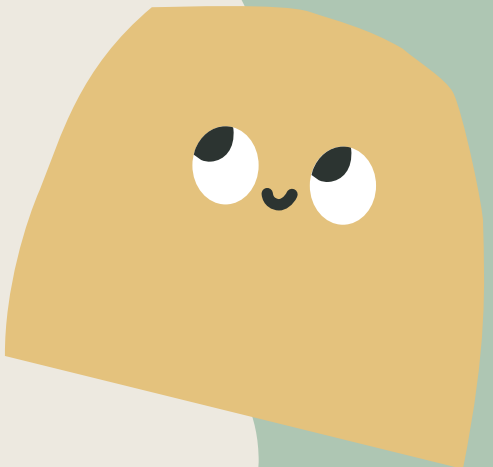
Conditional Statements

Lecture 5





Learning Objectives

➢By the end of this lecture, students should be able to:

- Understand decision-making in programming.
- Use if statements.
- Use if-else statements.
- Use multiple conditions with if-elif-else.
- Combine conditions using and, or, and not.
- Write nested conditional statements.
- Use comparison operators inside conditions.
- Build practical decision-making programs.
- Understand how conditional logic relates to AI agents.

Prepared By: Ehtisham Rasheed

Decision Making in Programming

- Imagine a student has obtained 65 marks
- We want the computer to decide:

```
If marks >= 50
    Student passed
```

- This is decision
- Another example

```
If temperature > 30
    It's hot
```

Prepared By: Engineer Ibrahim Rasheed

The if Statement

- The simplest conditional statement is if.
- Syntax:

```
if condition:
    statement
```

- Example:

```
age = 20

if age >= 18:
    print("You are an adult.")
```

Prepared By: Eng. Esham Rasheed

if-else

- The else block runs when the if condition is false
- Syntax:

```
if condition:
    statement
else:
    statement
```

- Example:

```
age = 16

if age >= 18:
    print("Adult")
else:
    print("Minor")
```

Prepared By: Ehtesham Rasheed

Examples

```python
number = int(input("Enter a number: "))

if number % 2 == 0:
    print("Even number")
else:
    print("Odd number")
```

```python
marks = float(input("Enter your marks: "))

if marks >= 50:
    print("Pass")
else:
    print("Fail")
```

```python
number = float(input("Enter a number: "))

if number >= 0:
    print("Positive")
else:
    print("Negative")
```

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if-elif-else

- Sometimes we have **more than two possibilities**
- For example:

```
Marks >= 80 → A
Marks >= 70 → B
Marks >= 60 → C
Marks >= 50 → D
Below 50 → F
```

- We can use:

```
if
elif
else
```

Prepared By: Engr. Basham Rasheed

if-elif-else

➢Syntax:

```
if condition1:
    statement
elif condition2:
    statement
elif condition3:
    statement
else:
    statement
```

Prepared by Ibrahim Rasheed

Grade Calculator

```python
marks = float(input("Enter your marks: "))

if marks >= 80:
    print("Grade A")
elif marks >= 70:
    print("Grade B")
elif marks >= 60:
    print("Grade C")
elif marks >= 50:
    print("Grade D")
else:
    print("Grade F")
```

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Multiple Conditions Using and, or, not

- We can combine conditions
- Suppose a student must have:
  - marks >= 50
  - attendance >= 75

```
marks = 65
attendance = 80

if marks >= 50 and attendance >= 75:
    print("Eligible")
else:
    print("Not eligible")
```

```
is_teacher = False
is_student = True

if is_teacher or is_student:
    print("University member")
```

```
is_raining = False

if not is_raining:
    print("You can go outside.")
```

Nested Conditions

- A nested condition means:
  - An **if** statement inside another **if** statement.
- Example:

```
age = 20
has_id = True

if age >= 18:
    if has_id:
        print("Entry allowed")
```

```
marks = 70
interview = 65

if marks >= 50:
    if interview >= 60:
        print("Admission successful")
```

```
if age >= 18 and has_id:
    print("Entry allowed")
```

Nested if-else

```
marks = 70

if marks >= 50:
    if marks >= 80:
        print("Excellent")
    else:
        print("Passed")
else:
    print("Failed")
```

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AI Connection — Intelligent Agent

- Imagine a simple robot.
- It receives:

```
battery = 15
obstacle = True
```

- The robot has to decide what to do

```
if battery < 20:
    print("Go to charging station")
else:
    if obstacle:
        print("Avoid obstacle")
    else:
        print("Move forward")
```

- This is a very simple representation of the basic idea behind an **intelligent agent**: Perceive → Decide → Act

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AI Example — Temperature Agent

➢Suppose an automated system receives room temperature

```
temperature = float(input("Enter temperature: "))

if temperature >= 30:
    print("Turn on air conditioner")
elif temperature >= 20:
    print("Maintain current temperature")
else:
    print("Turn on heater")
```

Practical Project — Student Grade System

```
name = input("Enter student name: ")
marks = float(input("Enter marks: "))

if marks >= 80:
    grade = "A"
elif marks >= 70:
    grade = "B"
elif marks >= 60:
    grade = "C"
elif marks >= 50:
    grade = "D"
else:
    grade = "F"

print(f"Student: {name}")
print(f"Marks: {marks}")
print(f"Grade: {grade}")
```

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Challenge 1 — Electricity Bill

- Take units consumed.
- Use the following simplified rules:

```
0–100 units       → Rs. 10/unit
101–200 units     → Rs. 15/unit
201–300 units     → Rs. 20/unit
Above 300         → Rs. 25/unit
```

- Display the estimated bill.

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Challenge 2 — University Admission Decision System

- Create a program that takes:
  - Student name
  - Matric/previous qualification percentage
  - Entry test score
  - Interview score
- Use these rules:

```
Previous percentage >= 60
AND
Entry test >= 50
```

- Then:
  - If interview >= 70 → **Strong Candidate**
  - If interview is 50–69 → **Candidate**
  - Below 50 → **Interview Requirement Not Met**
- If the basic eligibility requirements are not met, display: `Not Eligible`

Prepared By Mohammad Rasheed

Challenge 3 — Intelligent Temperature Controller

- Build a simulated intelligent temperature-control system.
- Inputs:
  - Current temperature
  - Desired temperature
  - Whether a person is present (yes/no)
- Rules:
  - If nobody is present → display "Energy Saving Mode“
  - If someone is present:
    - Current temperature > desired temperature + 2 → "Cooling ON“
    - Current temperature < desired temperature - 2 → "Heating ON“
    - Otherwise → "Temperature Comfortable"

Prepared By: Ehtisham Rasheed